

MOUNT GAMBIER, Australia

WMO: 948210

Lat: 37.7472S

Lon: 140.7739E

Elev: 226

StdP: 14.58

Time Zone: 9.50 (AUA)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	35.6	37.6	29.8	24.1	66.1	32.9	27.6	53.5	26.5	50.4	24.1	52.5	3.5	30	0.600

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	22.9	94.4	64.4	89.1	63.5	84.0	62.6	68.1	83.3	66.2	80.9	64.6	78.8	14.3	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.0	90.0	71.0	61.8	83.1	68.7	60.0	78.1	67.2	32.5	83.0	31.0	80.9	29.7	79.1	79.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.6	22.2	20.3	31.0	103.6	1.7	3.5	29.8	106.1	28.8	108.1	27.8	110.1	26.5	112.6		
(5)			31.0	72.5	1.5	2.6	30.0	74.3	29.1	75.8	28.3	77.2	27.2	79.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	56.6	65.0	65.7	62.3	57.5	53.4	49.8	49.8	52.0	54.7	58.6	61.8	(6)	
(7)		DBStd	7.99	7.10	6.98	6.50	5.27	4.24	3.43	3.17	3.68	4.39	5.81	6.62	6.69	(7)
(8)		HDD50	198	0	0	0	1	12	44	59	48	22	11	1	0	(8)
(9)		HDD65	3396	86	67	135	235	362	456	498	470	392	326	220	149	(9)
(10)		CDD50	2604	466	441	383	227	117	38	26	43	81	158	259	365	(10)
(11)		CDD65	326	87	88	52	11	1	0	0	0	1	9	28	49	(11)
(12)		CDH74	4304	1187	1027	631	137	6	0	0	0	6	130	427	753	(12)
(13)		CDH80	1982	609	497	277	35	0	0	0	0	0	37	167	360	(13)
(14)	Wind	WSAvg	10.2	11.0	10.8	9.9	8.7	8.9	9.1	10.2	10.6	10.8	10.6	10.5	10.9	(14)
(15)	Precipitation	PrecAvg	27.4	1.1	0.9	1.4	2.0	2.7	3.4	3.9	3.8	2.8	2.2	1.8	1.5	(15)
(16)		PrecMax	35.8	5.0	3.9	6.3	4.8	5.7	6.4	7.5	6.8	5.1	5.5	4.3	4.4	(16)
(17)		PrecMin	14.4	0.0	0.0	0.1	0.2	0.4	0.7	1.4	1.0	1.1	0.2	0.4	0.2	(17)
(18)		PrecStd	4.9	1.0	0.8	1.1	1.1	1.3	1.4	1.4	1.5	0.9	1.3	0.9	1.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	102.2	99.1	95.0	85.1	72.9	63.1	61.7	67.9	74.3	85.4	91.3	97.9	(19)
MCWB			65.3	66.1	63.6	60.1	56.3	55.3	53.1	54.3	58.0	60.5	63.3	62.9	(20)	
2%		DB	94.5	92.7	87.8	77.7	66.9	59.8	58.4	62.9	68.0	77.1	85.1	90.0	(21)	
		MCWB	64.9	64.8	63.1	58.8	56.3	53.8	51.9	52.8	55.8	58.5	62.7	62.7	(22)	
5%		DB	86.7	86.2	81.2	72.1	63.4	57.9	56.6	59.5	63.9	71.4	78.5	82.3	(23)	
		MCWB	63.6	64.3	61.3	57.8	55.6	52.8	51.1	52.0	54.2	57.2	61.4	62.1	(24)	
10%		DB	79.7	79.6	74.9	67.4	60.6	56.1	54.9	56.8	60.6	66.1	72.1	75.6	(25)	
		MCWB	62.6	63.2	60.6	56.9	54.8	51.7	50.3	50.8	52.7	55.7	59.6	60.4	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.9	71.1	67.8	63.5	60.2	57.5	55.6	56.4	60.4	63.3	68.4	68.4	(27)
MCD B			85.5	84.5	83.0	74.1	64.5	60.0	59.0	63.8	68.3	76.0	81.1	82.5	(28)	
2%		WB	68.1	67.5	65.3	61.4	58.4	55.3	53.5	54.3	57.4	60.3	65.1	65.6	(29)	
		MCD B	84.3	83.0	79.3	70.7	62.8	58.5	56.6	59.8	64.9	71.6	78.9	78.9	(30)	
5%		WB	65.8	65.8	63.4	59.7	57.0	53.7	52.2	52.9	55.4	58.4	62.4	63.4	(31)	
		MCD B	80.9	80.8	75.5	67.9	61.2	56.6	55.1	57.7	61.6	69.0	74.8	78.2	(32)	
10%		WB	63.6	63.9	61.6	58.1	55.6	52.4	51.1	51.6	53.7	56.3	60.0	61.2	(33)	
		MCD B	77.7	77.2	72.0	65.4	59.9	55.2	54.0	55.9	59.4	64.9	71.0	74.2	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	23.9	22.9	21.2	18.5	14.0	12.7	12.3	14.1	15.8	19.2	21.3	22.8	(35)
MCDBR			37.4	34.3	32.6	27.6	18.3	14.6	13.9	17.8	21.4	29.2	33.6	35.9	(36)	
MCWBR			12.2	11.3	11.8	12.1	10.7	10.1	9.5	10.6	12.1	14.1	14.3	13.4	(37)	
5% WB		MCDBR	29.7	28.1	25.9	21.2	14.5	12.7	12.3	15.6	18.2	25.3	29.1	29.6	(38)	
		MCWBR	12.3	11.1	11.3	11.1	9.6	9.6	9.2	10.1	11.4	13.6	14.0	13.0	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.336	0.335	0.326	0.315	0.309	0.300	0.300	0.317	0.336	0.341	0.338	0.337	(40)	
taud		2.524	2.539	2.561	2.578	2.570	2.586	2.582	2.521	2.457	2.447	2.483	2.506	(41)		
Ebn at Noon		316	309	299	284	265	259	266	278	291	304	314	318	(42)		
Edh at Noon		35	33	30	26	23	20	22	27	32	36	36	36	(43)		
(44)	All-Sky Solar Radiation	RadAvg	2328	2002	1571	1084	722	585	633	910	1281	1713	2045	2282	(44)	
(45)		RadStd	125	101	72	71	38	40	34	46	101	129	121	146	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.67	N/A	N/A	N/A	N/A	+0.88	N/A	-207	+248	+72
(47) Regional (22 neighbors)		+0.67	N/A	N/A	N/A	N/A	N/A	N/A	-173	+238	+70

Nomenclature: See separate page